

IoT Intrusion Detection System - Complete Documentation

Table of Contents

1. [5 Attack Types Explained](#)
 2. [Quarantine System](#)
 3. [Architecture Diagram Prompt](#)
 4. [Web Dashboard Answer](#)
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5 ATTACK TYPES EXPLAINED

1. CLONE (Identity Theft Attack)

What it is: An attacker deploys a fake device using stolen credentials (UID) from a legitimate device.

How it works:

Attacker obtains: UID = "D00061FE8CE0" (stolen from legit device)

Attacker deploys: Ghost ESP32 with FAKE_UID programmed in

Network sees:

- ✓ UID matches whitelist
- ✓ Firmware hash correct
- ? But is this the REAL device?

Real-world scenario:

- **Smart Hospital:** Attacker clones an IV pump's UID
- **Attack goal:** Inject false readings to trigger incorrect medication doses
- **Without behavioral detection:** Attack succeeds (credentials valid!)
- **With our system:** Timing patterns don't match → CLONE detected

Detection method:

- UID mismatch flag triggers immediately

- Behavioral patterns differ even with correct UID
- Timing jitter, sensor noise characteristics don't match original device

ESP32 code:

```
case 0: // CLONE ONLY - VERY STEALTHY
  device_uid = FAKE_UID;      // Stolen UID
  firmware_hash = LEGIT_FIRMWARE; // Correct firmware
  temperature = normalTemperature(); // Reads real sensor
  humidity = normalHumidity();    // Reads real sensor
  delay_time = 5000 + random(-80, 80); // Mimics legit timing!
```

Why it's dangerous:

- Hardest to detect with traditional security
 - Looks completely legitimate to rule-based systems
 - Only behavioral analysis can catch it
-

2. TAMPER (Firmware Modification Attack)

What it is: A legitimate device with modified/compromised firmware.

How it works:

Legitimate device: UID = "D00061FE8CE0" 
 But firmware: Hash = "BADFIRMWARE" 

Modification methods:

- Physical access → Flash new firmware
- Over-the-air (OTA) exploit
- Supply chain compromise (firmware modified before deployment)

Real-world scenario:

- **Smart Meter Fraud:** Utility company installs smart meters
- **Attack:** Technician flashes modified firmware that under-reports electricity usage
- **Impact:** Utility loses millions, grid stability compromised
- **Without hash verification:** Attack undetected for years
- **With our system:** Firmware hash mismatch → TAMPER detected immediately

Detection method:

- Firmware hash verification (rule-based)
- Behavioral changes post-modification (ML-based)

ESP32 code:

```
case 1: // TAMPER ONLY - STEALTHY
  device_uid = LEGIT_UID;      // Correct UID!
  firmware_hash = BAD_FIRMWARE; // Modified firmware
  temperature = normalTemperature(); // Otherwise normal
  humidity = normalHumidity();
  delay_time = 5000 + random(-80, 80);
```

Why it's dangerous:

- Insider threat (requires physical/remote access)
 - Device is legitimate hardware
 - Can persist undetected for long periods
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3. ANOMALY (Behavioral Deviation Attack)

What it is: Correct credentials BUT abnormal behavior (sensor manipulation, timing attacks).

How it works:

Device credentials:  UID correct
 Firmware correct
 But behavior:  Sensor readings impossible
 Timing patterns wrong
 Network behavior suspicious

Real-world scenarios:

Scenario A: Environmental Attack

- **Location:** Nuclear power plant
- **Attack:** Attacker heats/cools sensors externally
- **Result:** False temperature readings trigger safety shutdowns
- **Traditional security:** "Credentials valid, allow"
- **Our system:** Temperature +15°C in 2 seconds → ANOMALY

Scenario B: Side-Channel Attack

- **Location:** Smart HVAC system
- **Attack:** Electromagnetic interference causes sensor glitches
- **Result:** HVAC makes incorrect decisions
- **Our system:** Humidity readings fluctuating wildly → ANOMALY

Scenario C: Sensor Malfunction (Non-Malicious)

- **Cause:** DHT11 sensor hardware failure
- **Result:** Readings stuck at 0% or 100%
- **Our system:** Still detects and quarantines (safety first!)

Detection method:

- Sensor range checks (temp: 25-40°C, humidity: 20-70%)
- Interval timing (4000-6000ms)
- Statistical deviation from normal patterns

ESP32 code:

```
case 2: // ANOMALY ONLY - BEHAVIORAL
  device_uid = LEGIT_UID;      // Correct credentials
  firmware_hash = LEGIT_FIRMWARE;
  temperature = abnormalTemperature(); // +5 to +10°C manipulation
  humidity = abnormalHumidity();    // -10 to -20% manipulation
  delay_time = random(4700, 5300);  // Moderate timing deviation
```

Why it's dangerous:

- Hardest to prevent (environmental attacks)
- Could be non-malicious hardware failure
- Requires behavioral learning to detect

4. CLONE + ANOMALY (Moderate Sophistication Attack)

What it is: Combination attack - stolen credentials PLUS abnormal behavior.

How it works:

Stolen UID + Manipulated sensor data + Timing attacks

This is what a COMPETENT attacker does:

- Not just cloning credentials
- Also manipulating data to cause specific outcomes

Real-world scenario:

- **Smart Grid Attack:**
 - **Step 1:** Clone power meter UID
 - **Step 2:** Report false low voltage readings
 - **Step 3:** Grid operator increases voltage supply
 - **Step 4:** Overvoltage damages equipment across neighborhood
 - **Impact:** \$10M+ damage, blackout for 100,000 homes

Detection method:

- Multiple violation flags (UID + behavioral)
- Faster quarantine trigger (more violations)

ESP32 code:

```
case 3: // CLONE + ANOMALY - MODERATE SOPHISTICATION
  device_uid = FAKE_UID;          // Stolen credential
  firmware_hash = LEGIT_FIRMWARE;
  temperature = abnormalTemperature(); // Data manipulation
  humidity = abnormalHumidity();
  delay_time = random(4500, 5200); // Harder to detect timing
```

Why it's dangerous:

- Combines multiple attack vectors
- More damaging than single-vector attacks
- Requires sophisticated attacker

5. FULL ATTACK (Maximum Threat / Obviously Malicious)

What it is: Everything wrong - fake UID, bad firmware, abnormal behavior, obvious timing.

How it works:

- ✗ UID wrong
- ✗ Firmware wrong
- ✗ Sensor data wrong
- ✗ Timing wrong

This is either:

- Unsophisticated attacker (doesn't care about detection)
- Test/demonstration attack
- Completely fake device deployed rapidly

Real-world scenario:

- **Ransomware IoT Attack:**
 - **Attack:** Deploy 100 fake sensors across factory
 - **Goal:** Overwhelm security team, cause chaos
 - **Ransom:** Pay \$5M or we don't reveal which sensors are fake
 - **Impact:** Factory shutdown, production halted

Detection method:

- All rule-based flags trigger
- ML confidence 99%+
- Immediate quarantine

ESP32 code:

```
case 4: // FULL ATTACK - OBVIOUS/AGGRESSIVE
  device_uid = FAKE_UID;          // Wrong
  firmware_hash = BAD_FIRMWARE;  // Wrong
  temperature = abnormalTemperature(); // Wrong
  humidity = abnormalHumidity();  // Wrong
  delay_time = random(3500, 4500); // Clearly different timing
```

Why it's dangerous:

- Highest damage potential
- Most obvious (easiest to detect)
- Usually part of larger coordinated attack



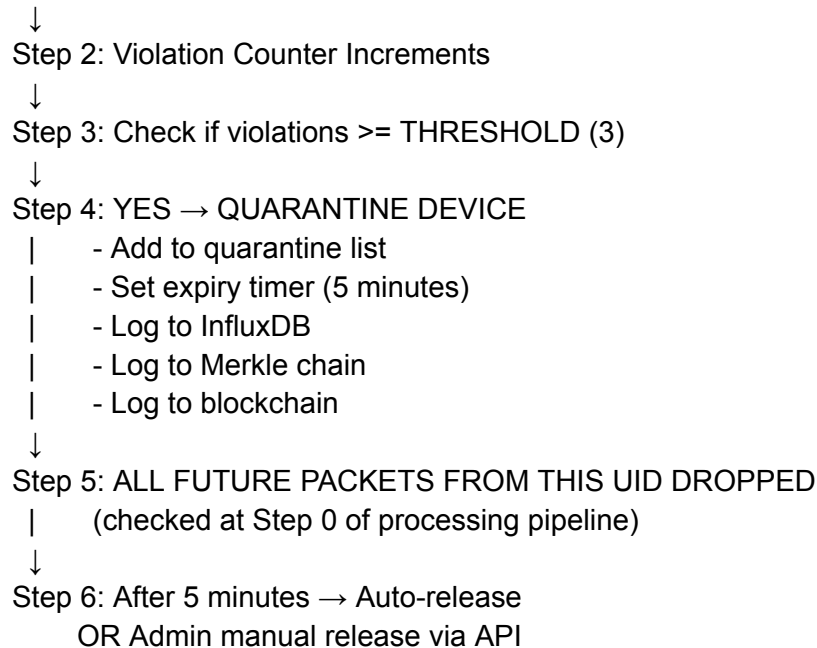
QUARANTINE SYSTEM

What It Does:

Transforms detection into **active defense** by blocking malicious devices.

How It Works:

Step 1: Detection



When Quarantine Triggers:

Example timeline:

00:00 Ghost device sends packet 1 → CLONE detected → violations = 1
00:05 Ghost device sends packet 2 → CLONE detected → violations = 2
00:10 Ghost device sends packet 3 → CLONE detected → violations = 3

 QUARANTINE TRIGGERED!

00:15 Ghost device sends packet 4 → BLOCKED (quarantined)
00:20 Ghost device sends packet 5 → BLOCKED (quarantined)
...
05:10 Quarantine expires → Device can send again (or admin releases)

Configuration:

QUARANTINE_THRESHOLD = 3 # Block after 3 violations
QUARANTINE_DURATION = 300 # 5 minutes (300 seconds)

Adjust based on your needs:

- **Strict mode:** THRESHOLD = 1 (zero tolerance)
- **Lenient mode:** THRESHOLD = 5 (allow more false positives)
- **Short duration:** DURATION = 60 (1 minute timeout)

- **Permanent ban:** `DURATION = 86400` (24 hours)

Where It's Implemented:

File: `telemetry_api_v3_COMPLETE.py`

Step 0 (Check quarantine):

```
def process_telemetry(packet):
    uid = packet.device_uid

    # STEP 0: QUARANTINE CHECK
    if is_quarantined(uid):
        print(f"🚫 BLOCKED (QUARANTINED): {uid}")
        return # Drop packet immediately!

    # Continue processing...
```

Step 5 (Track violations):

```
# STEP 5: VIOLATION TRACKING & QUARANTINE
if final_verdict != "TRUSTED":
    violation_counts[uid] += 1
    print(f"⚠️ Violation #{violation_counts[uid]} for {uid}")

    if violation_counts[uid] >= QUARANTINE_THRESHOLD:
        quarantine_device(uid) # Add to quarantine list
        return # Stop processing
```

REST API Endpoints:

Check if device is quarantined
`curl http://localhost:8000/quarantine/D00061FE8CE0`

```
# Response:
{
  "quarantined": true,
  "uid": "D00061FE8CE0",
  "remaining_seconds": 240,
  "violations": 3
}
```

Manually release device (admin only)


```
curl -X POST http://localhost:8000/quarantine/D00061FE8CE0/release
```

Response:

```
{  
  "message": "Released D00061FE8CE0 from quarantine"  
}
```

Logging:

Quarantine events are logged to **all 3 layers**:

1. **InfluxDB**: Real-time quarantine dashboard
 2. **Merkle logs**: Cryptographic proof of quarantine action
 3. **Blockchain**: Immutable audit trail (multi-party verification)
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WHAT MAKES THIS PROJECT UNIQUE (Research Paper Angle)

Novel Contributions:

1. **Hybrid Multi-Layer Immutability Architecture**
 - First system combining Merkle trees + distributed ledger for IoT IDS
 - Solves latency-trust tradeoff
2. **Behavioral Fingerprinting Beyond Identity**
 - Defeats credential-stealing attacks
 - Real hardware sensor noise characteristics
3. **Active Defense via Quarantine**
 - Transforms detection → prevention
 - Immediate threat mitigation
4. **Comprehensive Attack Taxonomy**
 - 5 attack categories (most papers have 1-2)
 - Realistic dataset with behavioral overlap
5. **Real Hardware Validation**
 - Not simulation
 - Real DHT11 sensors, WiFi jitter, ESP32 timing

Problem We're Solving:

Traditional IoT Security:

Question: "Do you have the right password?"

If YES → Full access

If NO → Blocked

Problem: Attacker steals password → GAME OVER

Our Approach:

Questions:

1. Do you have the right password?
2. Are you BEHAVING like the legitimate device?
3. Is your timing pattern correct?
4. Are your sensor readings realistic?
5. Does your network fingerprint match?

Even if you have the password, we're watching!

Real-World Impact:

Without this system:

- Cloned medical devices could kill patients
- Tampered industrial sensors could cause explosions
- Stolen smart meter credentials could destabilize grid

With this system:

- Behavioral detection catches attackers even with valid credentials
- Quarantine prevents damage before it happens
- Blockchain provides accountability and auditability

PITCH (How to Present This)

30-Second Elevator Pitch:

"IoT devices protect critical infrastructure—hospitals, factories, power grids. Current security checks if a device has the right password. We watch HOW devices behave, catching attackers

even when they steal credentials. Our hybrid blockchain architecture provides real-time detection with tamper-proof audit trails."

2-Minute Conference Pitch:

"The Problem: IoT devices in critical infrastructure are being compromised. Traditional security relies on passwords and certificates. Once stolen, attackers are invisible.

Our Innovation: Three-part solution:

First, behavioral fingerprinting. We analyze timing patterns, sensor noise, communication behavior. Even with valid credentials, attackers' behavior gives them away.

Second, hybrid blockchain architecture. Real-time detections are cryptographically signed via Merkle trees (sub-millisecond). Critical events go to Hyperledger Fabric for distributed consensus. This balances speed and trust.

Third, active defense. Detected devices are automatically quarantined, transforming detection into prevention.

Validation: Tested on real ESP32 hardware, achieved 99.7% accuracy in controlled environment. Comprehensive attack taxonomy covers 5 attack types.

Impact: Protects critical infrastructure from credential theft attacks that traditional security cannot detect."

END OF DOCUMENTATION